

PH6 / CRAM — CONSOLIDATED CANON STACK (v4.0)

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PREAMBLE: THE SIX-BOOK CANON

This document represents the fully consolidated, authoritative six-book canon stack for the PH6/CRAM deterministic scientific instrument. Prior structures, duplicate stacks, and four-document arrangements are permanently superseded.

The corpus is organized into six locked books. They must be read and interpreted in this exact sequence:

1. **Book 0 — Interpretive Control Plane:** Governs how doctrine is interpreted.
2. **Book I — Operational Source Constitution:** Governs what is true.
3. **Book II — Scientific Instrument Master:** Governs how truth is measured.
4. **Book III — Boundary Containment Annex:** Governs what is forbidden.
5. **Book IV — Certification Proof Pack:** Governs how truth is proven.
6. **Book V — Experimental Swarm Annex:** Governs what may be explored safely (Quarantined, Authority ZERO).

Master Ingestion Law

Interpret first. Constitute second. Measure third. Contain fourth. Prove fifth. Explore last. Never reverse this order.

Conflict Resolution Law

When sources conflict, resolve by:

1. Newest explicit instruction

2. Clearest technical source
3. Safest restrictive default

Never blend contradictions. Choose one source. Book doctrine overrides implementation. Certification verifies doctrine but may not rewrite it. Experimental material never outranks authority.

BOOK 0 — INTERPRETIVE CONTROL PLANE

Role: CANON | **Status:** ACTIVE **Question:** How doctrine is interpreted.

§ 0.1 System Identity Statement

PH6/CRAM is a deterministic scientific evidence instrument. It is not a general-purpose AI framework, not a generic software repository, and not an autonomous swarm architecture.

§ 0.2 Interpretive Precedence

Interpretations must follow the precedence ladder:

1. Newest explicit instruction
2. Clearest technical source
3. Safest restrictive default

§ 0.3 Controlled Vocabulary Lock

The following terms are locked and may not be redefined:

- **Engine:** Explanatory partition only. Never an autonomous authority.
- **Lane:** Authority boundary. Never an explanatory metaphor.
- **Layer:** Implementation decomposition. Never an authority source.
- **Mode:** Use “Framing” or “Posture” instead to avoid authority implications.
- **Scientific Instrument Framing:** Replaces the deprecated term “Scientifical”.

§ 0.4 Inference Firewall

Interpretation may not alter doctrine. Advisory material may not be merged into doctrine. Authority must not be inferred from metaphor. Implementation details must not be promoted into law.

§ 0.5 Drift Prohibition

Implementation may refine constraints but may never relax them. Any deviation breaks deterministic serialization and invalidates hash stability.

§ 0.6 Scientific Mode vs Authority Mode Separation

The system is strictly divided into two lanes. Lane 1 is the Authoritative Instrument Science lane, responsible for capture, preservation, measurement, and adjudication. Lane 2 is the Exploratory Scientific Analysis lane, responsible for hypothesis generation, experimentation, and advisory AI. The boundary between them is a One-Way Membrane: Lane 1 may seed Lane 2, but Lane 2 may never influence Lane 1.

§ 0.7 Interpretation Receipts

Every authoritative decision must produce a signed receipt with a BLAKE2b-256 hash. All interpretations must pass the Audit Gate before being accepted as CANON.

§ 0.8 Book 0 Lock Statement

Book 0 does not create authority. Book 0 governs access to authority.

§ 0.9 Constitutional Relationship Clause

Book 0 gates access to Book I. Book I is the doctrinal authority.

§ 0.10 Negative Doctrine / Forbidden Transformations

The following transformations are explicitly forbidden:

1. Promoting Advisory content to Authority.
2. Using Engine models as Verdict paths.

3. Allowing Scientific mode to create Doctrine exceptions.
4. Granting Swarm concepts Governance authority.
5. Allowing Lane 2 content to mutate Lane 1 state.

§ 0.11 Audit Gate Rule

All interpretations must pass the Audit Gate before being accepted as CANON. This is mandatory in Continuous Integration (CI).

§ 0.12 Open Issues Register

The following items remain open and tracked:

1. Sensor-specific re-derivation for non-optical sensors.
2. Final HRG9 seal procedures.
3. MRAM-R canonization.
4. WARM storage tier definition.
5. HotWriter production release.
6. Kubernetes-owned CRAM-A/0 governance.

§ 0.13 Interpretive Separation Law

Substance (Doctrine) is separate from the Interpretation Plane.

§ 0.14 Canonical Constraint Compiler

Refer to `policy.yaml` and `invariants.yaml` for compiled constraints.

§ 0.15 Recursive Drift Prohibition + Boundary Inheritance Law

Constraints may only be strengthened, never weakened. Any extension or patch must inherit the most restrictive containment boundaries of its parent.

§ 0.16 Scientific Mode Patch

The “Six-Engine” model is explanatory only. It does not grant autonomous authority to any engine.

§ 0.17 Integrated Precedence Ladder

1. DOC 0 (Constitution)
 2. Boundary Inheritance Law (BIL-1.0)
 3. CANON Documents (Books I-II)
 4. CERTIFICATION Artifacts (Book IV)
 5. ANNEX / EXPERIMENTAL (Book III/V)
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BOOK I — OPERATIONAL SOURCE CONSTITUTION

Role: CANON | **Status:** ACTIVE **Question:** What is true.

§ 1.1 System Identity and Governing Principle

PH6/CRAM is a deterministic scientific evidence instrument. The core seal is: Capture everything configured. Preserve it exactly. Measure deterministically. Interpret separately. Prove everything.

§ 1.2 Lane Architecture and Authority Model

The system operates on a strict dual-lane architecture. Lane 1 is authoritative, handling capture, preservation, measurement (PSEUDO-M), and adjudication (PSEUDO-A). Its authority is absolute. Lane 2 is advisory, handling hypothesis, experimentation, and advisory AI. Its authority is zero.

The PSEUDO Authority Invariant dictates that PSEUDO alone adjudicates. Scientific mode may not adjudicate. The Scientific Extension Invariant states that PSEUDO-SCI extends observables, never verdict authority. The Science Consumption Law allows science to consume PSEUDO outputs but forbids science from tuning PSEUDO thresholds.

§ 1.3 Seven Iron Laws

1. Determinism over intelligence.
2. Auditability over adaptability.
3. PSEUDO authority over AI.
4. Never Block RSYNC.
5. Offline equals Online.
6. Explicit modes only.
7. Bounded behavior always.

§ 1.4 PASS/DROP Exclusivity + Vocabulary Lock

The only valid verdict terms are PASS and DROP. Terms such as ACCEPT, REJECT, FLAG, HOLD, REVIEW, and RETAIN are strictly forbidden. PASS/DROP exclusivity belongs solely to PSEUDO-A in Lane 1.

§ 1.5 One-Way Membrane

The boundary between Lane 1 and Lane 2 is a one-way membrane. Lane 1 may seed Lane 2. Lane 2 may never influence Lane 1 in any form.

§ 1.6 Boundary Inheritance Law

Implementation may refine constraints but may never relax them. All extensions must inherit the most restrictive bounds of the parent.

§ 1.7 Update Exception Clause

Constitutional sources override all implementation, patch, and advisory content.

§ 1.8 FAST/SLOW Doctrine

Lane 1 execution must be FAST. Lane 2 execution is SLOW and advisory.

§ 1.9 Six-Engine Explanatory Mapping

The Six-Engine model is explanatory only and holds no authority:

- Engine I: PSEUDO (Authority)
- Engine II: Runtime (FAST)
- Engine III: Persistence (HOT)
- Engine IV: Topology (SoSo)
- Engine V: Swarm (Disabled)
- Engine VI: Instrument Science (PSEUDO-SCI)

§ 1.10 Swarm-Disabled Constitutional Constraints

Swarm is disabled in production and holds zero authority.

§ 1.11 Truth Class Separation

Observed Truth and Verdict Truth reside in Lane 1 with absolute authority. Hypothesis Truth resides in Lane 2 with zero authority.

§ 1.12 Constitutional Invariants

The following ten invariants are locked:

1. RSYNC Sovereignty
2. Atomic Write Contract
3. CRAM Immutability
4. Deterministic Hashing (BLAKE2b-256)
5. Canonical JSON Serialization
6. Lane Isolation
7. Verdict Exclusivity
8. No Runtime Learning in Lane 1
9. Crash Consistency
10. Audit Traceability

§ 1.13 AI Reconstruction Rules

AI may be used for reconstruction in Lane 2 but has zero authority over Lane 1 evidence.

§ 1.14 Functional Partition Table

- Partition I: Measurement/Adjudication (Sole Lane-1 Verdict Authority)
- Partition II: Runtime (None)
- Partition III: Persistence (None)
- Partition IV: Topology (Advisory only)
- Partition V: Distributed Support (Disabled)
- Partition VI: Instrument Science (Advisory only)

§ 1.15 Swarm Constitutional Containment Rule

Swarm outputs must be confined to MRAM-S.

§ 1.16 Framing Rule

There is no authority difference between different scientific framings.

§ 1.17 No Authority Widening Rule

It is forbidden to derive authority from partitions, create alternate PASS/DROP routes, or allow advisory systems to influence thresholds.

§ 1.18 DRIFT_FAIL Condition Definitions

A DRIFT_FAIL condition occurs if Lane 2 writes to CRAM, AI emits a verdict, a verdict is not PASS/DROP, or thresholds mutate at runtime.

§ 1.19 Engine VI Constitutional Limits and Boundary Laws

Engine VI may advise science but may not govern authority.

BOOK II — SCIENTIFIC INSTRUMENT MASTER

Role: CANON | **Status:** ACTIVE **Question:** How truth is measured.

§ 2.1 Deterministic Instrument Science Objectives and Scope

The objectives of the scientific instrument are preservation, determinism, crash safety, auditability, scientific use, authority isolation, and export sovereignty.

§ 2.2 Six-Engine Runtime Model

The runtime model consists of Engine I (PSEUDO) for deterministic authority, Engine II (Runtime) for FAST execution, Engine III (Persistence) for HOT append-only storage, Engine IV (Topology) for SoSo advisory, Engine V (Swarm) which is runtime blocked, and Engine VI (Instrument Science) for PSEUDO-SCI sideband operations.

§ 2.3 Calibration Doctrine

Deterministic metrics must be calibrated to sensor-specific baselines without allowing runtime adaptation.

§ 2.4 CRAM Storage Model

The storage model includes CRAM-0 as a write-first intake buffer, CRAM-A for immutable authoritative evidence (PASS), CRAM-R as a reject vault (DROP), and MRAM-S for advisory shadow storage (Lane 2).

§ 2.5 Sensor Normalization

All sensor inputs must be normalized to deterministic fixed-point representations before PSEUDO-M processing.

§ 2.6 `evaluate_frame` Contract

The `evaluate_frame` contract must return a verdict (PASS/DROP), fixed-point metrics, reasons, input hash, gate profile version, and threshold profile version.

§ 2.7 Entropy / Laplacian / Motion Gates

The deterministic gates are defined as follows:

- Entropy: $3.5 \leq H \leq 7.5$
- Laplacian Variance: $\text{Var} \geq 60.0$
- Motion Fraction: $0.05 \leq M \leq 0.95$

§ 2.8 PSEUDO Measurement Doctrine

PSEUDO-M measures Shannon Entropy, Laplacian Variance, and Motion Fraction. PSEUDO-A adjudicates based on fixed thresholds. Any veto results in a DROP verdict.

§ 2.9 EvidencePacket and Schema Definitions

An EvidencePacket is a sealed boundary object containing the CRAM object ID, source ID, UTC timestamp, sequence number, SHA-256 hash, size in bytes, CRAM tier, gate scores, verdict, veto reasons, policy version, and a canon lock flag.

§ 2.10 Absolute Constraints

There is no AI verdict authority, no runtime learning in Lane 1, and no adaptive threshold drift.

§ 2.11 Atomic Write Contract

The atomic write sequence consists of four steps:

1. Write raw data to CRAM-0.
2. Compute metrics and verdict.
3. Write EvidencePacket to CRAM-A/R.
4. Update authority chain hash.

§ 2.12 Standard Operating Procedures

Procedures are defined for intake, preservation, and export via RSYNC.

§ 2.13 Authoritative Science Lane (Lane 1)

Lane 1 is the primary path for evidence generation.

§ 2.14 Exploratory Science Lane (Lane 2)

Lane 2 is the path for hypothesis and advisory analysis.

§ 2.15 Recovery and Debugging

Crash-consistency ensures that power loss does not create partial authoritative state.

BOOK III — BOUNDARY CONTAINMENT ANNEX

Role: CANON | **Status:** ACTIVE **Question:** What is forbidden.

§ 3.1 Containment Doctrine

The containment layer ensures that advisory systems (Lane 2) never contaminate the authoritative instrument (Lane 1).

§ 3.2 Swarm Prohibitions

Swarm is DISABLED in production and has ZERO authority. Swarm may not write to CRAM-0, CRAM-A, or CRAM-R, and may not influence PASS/DROP verdicts.

§ 3.3 AI / SoSo Containment

AI outputs are advisory only. SoSo topology is confined to MRAM-S. No AI-generated content may enter an EvidencePacket.

§ 3.4 RSYNC Protection

Advisory services must never block or delay RSYNC export. This is Priority Zero.

§ 3.5 Forbidden Transformations

The following transformations are strictly forbidden:

- Promoting Advisory to Authority.
- Using Engine models as Verdict paths.
- Allowing Scientific mode to create Doctrine exceptions.

§ 3.6 Boundary Inheritance Law (Containment)

Any extension or patch must inherit the most restrictive containment boundaries.

§ 3.7 Drift Detection and DRIFT_FAIL

The system must emit a DRIFT_FAIL and halt if any containment boundary is breached.

BOOK IV — CERTIFICATION PROOF PACK

Role: CERTIFICATION | **Status:** ACTIVE **Question:** How truth is proven.

§ 4.1 Certification Law

Certification verifies doctrine; certification does not modify doctrine. Passing certification demonstrates conformance but does not redefine doctrine. Failure indicates implementation deviation, not doctrinal mutation.

§ 4.2 Authority Limits

Certification may verify implementation, certify conformance, and detect drift. Certification may not change doctrine, widen authority, override constitutional sources, convert certification into governance, reinterpret doctrine via test design, alter thresholds by implication, create authority through tests, or promote advisory outputs into evidence.

§ 4.3 Validation Principles

Validation confirms schema integrity, replay determinism, boundary compliance, and certification conformance.

§ 4.4 Schema Contracts

The EvidencePacket schema is a frozen dataclass that accepts only PASS or DROP verdicts. The canon lock must be true, and the CRAM tier must be CRAM-A or CRAM-R. No advisory fields may appear in this structure.

The Receipt Schema (ph6_receipt_v1) requires a monotonically increasing event sequence, an event hash (BLAKE2b-256 of event content), and an authority hash (BLAKE2b-256 of authority chain).

The Replay Manifest Schema (ph6_replay_v1) requires identical verdict behavior, schema stability, canonical serialization, and evidence boundary isolation.

The MRAM-S Advisory Schema (ph6_mram_s_advisory_v1) requires the authority level to always be “ZERO”. No MRAM-S artifact may influence Lane 1.

The Audit Event Schema (ph6_audit_v1) requires all eight fields to be emitted. Any audit implementation missing the event sequence or authority hash is non-conformant and must be patched before certification.

§ 4.5 Hash and Serialization Doctrine

The primary evidentiary hash is BLAKE2b-256. Canonical JSON rules require sorted keys, no ASCII enforcement, and no NaN values.

§ 4.6 C1 Golden Vectors

The canonical determinism conformance test set requires the verdict, metrics, and result set hash to match exactly.

§ 4.7 HRG9 Evidence Gate

The HRG9 Evidence Gate status is OPEN / EVIDENCE BLOCKED. This is a STOP-SHIP condition. Closure requires P1/P2/P3 vector parity PASS, result set hash parity across

all three, authority hash verification PASS, the presence of the HRG9 manifest, and the emission of the HRG9 replay parity receipt.

§ 4.8 Certification Matrix

The certification matrix includes EvidencePacket schema conformance, replay determinism proof, BLAKE2b-256 hash stability, canonical JSON conformance, atomic write power-loss safety, lane boundary isolation, and HRG9 evidence gate closure.

BOOK V — EXPERIMENTAL SWARM ANNEX

Role: EXPERIMENTAL | **Status:** ACTIVE (Advisory and research use only) **Question:** What may be explored safely. **Authority:** ZERO. Lane 2 only. MRAM-S only. Never in authority stack.

§ 5.1 Advisory Boundary Notice

This document is advisory research material only. It has ZERO authority and operates exclusively in Lane 2 and MRAM-S.

§ 5.2 L2_EXPAND Scientific Advisory Token Mode

This mode enables bounded scientific advisory token analysis without altering deterministic authority.

§ 5.3 Swarm Token Science Model (STSM-1.0)

The Swarm Token Science Model is a Lane-2-only research model in which SoSo, AI, and Swarm workers generate, compare, decay, stabilize, and organize advisory tokens into a non-evidentiary hypothesis topology stored only in MRAM-S. It has Authority ZERO and no reverse path to Lane-1 measurement, adjudication, evidence, CRAM-A, EvidencePacket, replay, or PASS/DROP.

The token classes are:

- **RT (Reference Token):** Anchors advisory work to Evidence References. Authority: Zero.

- **VDT (Virtual Decay Token):** Temporary continuity / decay hypothesis. Authority: Zero.
- **VLT (Virtual Longevity Token):** Stabilized advisory continuity token. Authority: Zero.

§ 5.4 Token Science Pipeline

The pipeline flows from Lane 1 Authoritative Evidence (seed only) to Evidence References, Reference Token Generation, Swarm Token Workers, Advisory Token Comparison, Decay / Stability Analysis, VDT, VLT Advisory Promotion, Advisory Hypothesis Topology, and finally MRAM-S Storage Only.

§ 5.5 Swarm Worker Roles

- **Reference Worker:** Builds RTs from Evidence References.
- **Decay Worker:** Models token weakening over time.
- **Comparison Worker:** Compares competing token paths.
- **Stability Worker:** Tests persistence across windows.
- **Topology Worker:** Builds advisory hypothesis graph.
- **Audit Worker:** Confirms no authority contamination.

§ 5.6 Safety Constraints

The following actions are allowed: Lane 1 may seed Lane 2; Swarm may compare advisory tokens, generate scientific hypotheses, promote VDT to VLT inside advisory policy, and store results in MRAM-S.

The following actions are strictly forbidden: CRAM mutation, CRAM-A write, PASS/DROP influence, EvidencePacket inclusion, authority-chain participation, replay dependency, authority return, and authority derivation.

GAP REGISTER

The following gaps remain unresolved and are explicitly preserved. They must not be hidden or falsely closed.

Book II Gaps

- **GAP-B2-1:** MRAM-S advisory result schema is confirmed missing in the code pack. This is an audit issue requiring schema authorship before certification.
- **GAP-B2-2:** Scientific experiment metadata schema is confirmed missing in the code pack. This is an audit issue requiring schema authorship.

Book III Gaps

- **GAP-B3-1:** SoSo isolation extraction lacks a dedicated standalone source. It must be extracted from the instrument master and formalized in Book III § 3.
- **GAP-B3-2:** Forbidden transformations are scattered across multiple patch PDFs. A full consolidated list must be compiled in Book III § 5.

Book IV Gaps (STOP-SHIP Conditions)

- **GAP-B4-1:** HRG9 Evidence Gate is OPEN / EVIDENCE BLOCKED. This is a critical STOP-SHIP condition. Production release is blocked until closed.
- **GAP-B4-2:** The `audit.py` authority hash and event sequence defect is a confirmed open defect. The `append_audit()` function must be patched to emit all 8 mandatory fields before certification.
- **GAP-B4-3:** Resource-cage compliance test population is flagged as needing population.
- **GAP-B4-4:** Drift Gate Specification requires verification.

Book V Gaps

- **GAP-B5-1:** Swarm Mesh extension stubs are referenced but not confirmed in available PDFs.
 - **GAP-B5-2:** Simulation/deception annexes are referenced as being from non-uploaded sources.
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RETIREMENT QUEUE

The following sources have been absorbed into the canonical homes and are retired to the `SUPERSEDED/` directory:

1. `PH6_CRAM__SCIENTIFIC_INSTRUMENT_AND_SCIENTIFIC_MODE_MASTER.pdf` (Superseded by v1.1)
2. `PH6_CRAM__SIX_ENGINE_SCIENTIFIC_MODE.pdf` (Absorbed into Book II § 2)
3. `SIX_ENGINE_SWARM_ADVISORY_SOURCE.pdf` (Split between Book III containment and Book V advisory)
4. `03_BOOK_IV_ADDENDUM_A_CCA_1.1.pdf` (Absorbed into Book I § 15-21)
5. `PSEUDO_EXPLICIT_BOUNDARY_PATCH.pdf` (Absorbed into Book I § 2 and Book II § 2)
6. `SIX_ENGINE_SUBJECT_MATTER_UPDATE_FRAME.pdf` (Absorbed into Book I § 22 and Book II § 2)
7. `01_PH6_OPERATIONAL_SOURCE_CANON_v2.0.pdf` (Split between Book I constitutional and Book II implementation)
8. `__PH6_BOOK_0__INTERPRETIVE_CONTROL_PLANE_LOCK_PATCH.pdf` (Merged into Book 0 § 8-13)
9. `BOOK_0_CONSOLIDATED_CANON_PATCH.pdf` (Merged into Book 0 § 14-19)
10. `README-1.pdf` (Four-doc stack replaced by six-book structure)

The following sources remain active in the `EXPERIMENTAL/` directory:

- `Swarm_tokens_.pdf` (Book V base)
- `05_BOOK_V_EXPERIMENTAL_SWARM_ANNEX.pdf` (Book V canonical)